
NUMERICAL COMPUTATION OF HADAMARD FINITE PART OF SOME HYPERSINGULAR INTEGRALS IN 1, 2 AND 3 DIMENSIONS

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September 16, 2026

ABSTRACT

1 Introduction

2 Notations

For every physical quantity, the corresponding reference space quantity is denoted by a hat, so that, for example, if $\mathbf{x}$ is a point on a d -dimensional variety, then $\hat{\mathbf{x}}$ is the corresponding point in the reference space. If u is a function defined on the variety, then $\hat{u}$ is the corresponding function in the reference space. Furthermore, we denote by $\oint$ the Hadamard finite part of an integral.

COMBINATORY NOTATIONS: as in the article [[constantine_multivariate_1996](#)], we will use the following notations for multi-indices and combinatory numbers. If $\boldsymbol{\nu} = (\nu_1, \dots, \nu_r) \in \mathbb{N}_0^r$ and $\mathbf{z} = (z_1, \dots, z_r) \in \mathbb{R}^r$, then

$$|\boldsymbol{\nu}| = \sum_{i=1}^r \nu_i, \quad \boldsymbol{\nu}! = \prod_{i=1}^r \nu_i!, \quad \mathbf{z}^{\boldsymbol{\nu}} = \prod_{i=1}^r z_i^{\nu_i}, \quad \partial_{\mathbf{x}}^{\boldsymbol{\nu}} = \frac{\partial^{|\boldsymbol{\nu}|}}{\partial x_1^{\nu_1} \dots \partial x_r^{\nu_r}} \quad (1)$$

as well as the following combinatory sets:

$$p(n, k) = \left\{ (\lambda_1, \dots, \lambda_k) : \lambda_i \in \mathbb{N}_0, \sum_{i=1}^k \lambda_i = k, \sum_{i=1}^k i \lambda_i = n \right\} \quad (2)$$

and its partially vectorized counterparts:

$$p(\boldsymbol{\nu}, r) = \left\{ (k_1, \dots, k_n; \ell_1, \dots, \ell_n) : \exists 1 \leq s \leq n : \forall 1 \leq i \leq n-s, \right. \\ \left. k_i = 0 \cap \ell_i = \mathbf{0}, \quad \forall n-s+1 \leq i \leq n, k_i > 0, \right. \\ \left. \text{et } \mathbf{0} \prec \ell_1 \prec \dots \prec \ell_n \text{ are such that} \right. \\ \left. \sum_{i=1}^n k_i \ell_i = \boldsymbol{\nu}, \quad \sum_{i=1}^n k_i = r \right\}, \quad n = |\boldsymbol{\nu}| \quad (3)$$

$$p(n, \boldsymbol{\lambda}) = \left\{ (k_1, \dots, k_n) : \mathbf{k}_j \leq 0, \sum_{i=1}^n \mathbf{k}_i = \boldsymbol{\lambda}, \sum_{i=1}^n i|\mathbf{k}_i| = n \right\} \quad (4)$$

Finally, the fully vectorized combinatory set is defined as

$$p_s(\boldsymbol{\nu}, \boldsymbol{\lambda}) = \left\{ ((\mathbf{k}_1, \dots, \mathbf{k}_s); (\ell_1, \dots, \ell_s)) : |\mathbf{k}_i| > 0, \mathbf{0} \prec \ell_1 \prec \dots \prec \ell_s, \sum_{i=1}^s \mathbf{k}_i = \boldsymbol{\lambda} \text{ and } \sum_{i=1}^s |\mathbf{k}_i| \ell_i = \boldsymbol{\nu} \right\} \quad (5)$$

For each of these set, we let the following quantity be defined:

$$P_{\boldsymbol{\nu}, r} = \sum_{p(\boldsymbol{\nu}, r)} \frac{1}{\prod_{i=1}^n k_i! \ell_i!^{k_i}}, \quad P_{n, \boldsymbol{\lambda}} = \sum_{p(n, \boldsymbol{\lambda})} \frac{1}{\prod_{i=1}^n k_i! i!^{|\mathbf{k}_i|}}, \quad P_{\boldsymbol{\nu}, \boldsymbol{\lambda}}^s = \sum_{p_s(\boldsymbol{\nu}, \boldsymbol{\lambda})} \frac{1}{\prod_{i=1}^s \mathbf{k}_i! \ell_i!^{|\mathbf{k}_i|}} \quad (6)$$

The bold notation will indicate which combinatory set is looked at.

3 Model presentation

Let's consider an finite part integral of the form

$$I = \oint_{\hat{\tau}} f(\hat{\mathbf{y}}) d\hat{\mathbf{y}}, \quad f(\hat{\mathbf{y}}) \underset{\hat{\mathbf{y}} \rightarrow \hat{\mathbf{x}}}{\sim} \frac{1}{\|\hat{\mathbf{y}} - \hat{\mathbf{x}}\|^{d+1}}. \quad (7)$$

4 Richardson extrapolation for the computation of the Laurent's coefficients

4.1 General idea

For any angle θ , let's consider a finite sequence of n radial points $\rho_i(\theta)$ such that

$$0 < \rho_1(\theta) < \rho_2(\theta) < \dots < \rho_n(\theta) < \hat{\rho}(\theta), \quad (8)$$

Recall: we can express the two first terms of the Laurent expansion of the complete polar kernel $F(\rho, \theta) = \rho K(\chi(\hat{\mathbf{x}}), \chi(\hat{\mathbf{y}}(\rho, \theta))) J(\hat{\mathbf{y}}(\rho, \theta)) \hat{u}(\hat{\mathbf{y}}(\rho, \theta))$ as the following limits

$$F_{-2}(\theta) = \lim_{\rho \rightarrow 0} \rho^2 F(\rho, \theta), \quad F_{-1}(\theta) = \lim_{\rho \rightarrow 0} \rho \left(F(\rho, \theta) - \frac{F_{-2}(\theta)}{\rho^2} \right). \quad (9)$$

such that we can write the complete polar kernel as

$$F(\rho, \theta) = \frac{F_{-2}(\theta)}{\rho^2} + \frac{F_{-1}(\theta)}{\rho} + O_{\rho \rightarrow 1}(1) \quad (10)$$

Let $c_{1,i}$ and $c_{2,i}$ be the coefficients used in the Richardson extrapolation of order n to compute the previous limits, so that we can write

$$\boxed{F_{-2}(\theta) \approx \tilde{F}_{-2}(\theta) = \sum_{i=1}^n c_{2,i} \rho_i^2(\theta) F(\rho_i(\theta), \theta)}, \quad F_{-1}(\theta) \approx \tilde{F}_{-1}(\theta) = \sum_{i=1}^n c_{1,i} \rho_i(\theta) \left(F(\rho_i(\theta), \theta) - \frac{\tilde{F}_{-2}(\theta)}{\rho_i^2(\theta)} \right). \quad (11)$$

By injecting the formula of $\tilde{F}_{-2}(\theta)$ into the formula of $\tilde{F}_{-1}(\theta)$, so that we can write $\tilde{F}_{-1}(\theta)$ only as a linear combination of the complete polar kernel evaluated at the radial points $\rho_i(\theta)$, we obtain

$$\tilde{F}_{-1}(\theta) = \sum_{i=1}^n c_{1,i} \rho_i(\theta) F(\rho_i(\theta), \theta) - \sum_{i=1}^n c_{1,i} \frac{1}{\rho_i(\theta)} \sum_{j=1}^n c_{2,j} \rho_j^2(\theta) F(\rho_j(\theta), \theta).$$

By letting $P_n(\theta) = \sum_{k=1}^n \frac{1}{\rho_k(\theta)}$, we can write the previous formula as

$$\boxed{\tilde{F}_{-1}(\theta) = \sum_{i=1}^n F(\rho_i(\theta), \theta) \rho_i(\theta) \left(c_{1,i} - c_{2,i} P_n(\theta) \rho_i(\theta) \right)} \quad (12)$$

4.2 Lagrange basis method

For this section and the following, we will no longer use θ in the notation, it will be implicit. Let's denote the regular function $G(\rho) = \rho^2 F(\rho)$, so that $G(0) = F_{-2}$ and $G'(0) = F_{-1}$. Let $\ell_i(\rho)$ be the Lagrange basis of degree $n-1$ associated to the radial points ρ_i , defined as

$$\ell_i(\rho) = \prod_{j \neq i} \frac{\rho - \rho_j}{\rho_i - \rho_j}, \quad i = 1, \dots, n. \quad (13)$$

The lagrange interpolant of G and its derivative are given by

$$\Pi_{n-1} G(\rho) = \sum_{i=1}^n G(\rho_i) \ell_i(\rho), \quad (\Pi_{n-1} G)'(\rho) = \sum_{i=1}^n G(\rho_i) \ell'_i(\rho). \quad (14)$$

So that the numerical Laurent's coefficients, denoted $\tilde{F}_{-2}$ and $\tilde{F}_{-1}$, are simply the evaluation of the interpolant and its derivative at the origin $\rho = 0$:

$$\tilde{F}_{-2} = \Pi_{n-1} G(0) = \sum_{i=1}^n G(\rho_i) \ell_i(0), \quad \tilde{F}_{-1} = (\Pi_{n-1} G)'(0) = \sum_{i=1}^n G(\rho_i) \ell'_i(0). \quad (15)$$

with $\ell_i(0) = \prod_{j \neq i} \frac{\rho_j}{\rho_j - \rho_i}$ and $\ell'_i(0) = \ell_i(0) \left(\frac{1}{\rho_i} - P_n \right)$, so that we can finally write the Laurent's coefficients as

$$\tilde{F}_{-2} = \sum_{i=1}^n \ell_i(0) \rho_i^2 F(\rho_i), \quad \tilde{F}_{-1} = \sum_{i=1}^n \ell_i(0) \rho_i (1 - P_n \rho_i) F(\rho_i). \quad (16)$$

This is a particular case of the Richardson extrapolation, with the two families of coefficients $c_{1,i}$ and $c_{2,i}$ collapsing into a single family of coefficients $\ell_i(0)$:

$$c_{1,i} = c_{2,i} = \ell_i(0) = c_i = \prod_{j \neq i} \frac{\rho_j}{\rho_j - \rho_i}.$$

Finally, the numerical Laurent's coefficients can be written as

$$\tilde{F}_{-2}(\theta) = \sum_{i=1}^n A_i(\theta) F(\rho_i(\theta), \theta), \quad \tilde{F}_{-1}(\theta) = \sum_{i=1}^n B_i(\theta) F(\rho_i(\theta), \theta). \quad (17)$$

With $A_i(\theta) = c_i(\theta)\rho_i^2(\theta)$ and $B_i(\theta) = c_i(\theta)\rho_i(\theta)(1 - P_n(\theta)\rho_i(\theta))$.

Each of these formula simply consist in the numerical evaluation of the polar kernel at the radial points $\rho_i(\theta)$, multiplied by some weights.

Remark 4.1. Properties of $A_i(\theta)$ and $B_i(\theta)$

$$\sum_{i=1}^n A_i = \begin{cases} \rho_1, & n = 1 \\ -\rho_1\rho_2, & n = 2 \\ 0, & n \geq 3 \end{cases}, \quad \sum_{i=1}^n B_i = \begin{cases} 0, & n = 1 \\ \rho_1 + \rho_2, & n = 2 \\ 0, & n \geq 3 \end{cases} \quad (18)$$

These three cases illustrates the following definition :

$$\sum_{i=1}^n A_i = (\Pi_{n-1}[\rho^2])(0), \quad \sum_{i=1}^n B_i = (\Pi_{n-1}[\rho^2])'(0) \quad (19)$$

4.3 Final quadrature rule

Let's then denote $I(\theta)$ the integral of the complete polar kernel over the radial direction, defined as:

$$I(\theta) = \int_0^{\hat{\rho}(\theta)} \left\{ F(\rho, \theta) - \frac{F_{-1}(\theta)}{\rho} - \frac{F_{-2}(\theta)}{\rho^2} \right\} d\rho + F_{-1}(\theta) \ln \left| \frac{\hat{\rho}(\theta)}{\beta(\theta)} \right| - F_{-2}(\theta) \left\{ \frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)} \right\} \quad (20)$$

A final quadrature rule for it is the following:

$$I(\theta) \approx \tilde{I}(\theta) = \sum_{i=1}^n \tilde{w}_i(\theta) F(\rho_i(\theta), \theta) \quad (21)$$

With the new weights $\tilde{w}_i(\theta)$ defined as

$$\tilde{w}_i(\theta) = w_i(\theta) - A_i(\theta) \left\{ W_{-2,n}(\theta) + \frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)} \right\} + B_i(\theta) \left\{ \ln \left| \frac{\hat{\rho}(\theta)}{\beta(\theta)} \right| - W_{-1,n}(\theta) \right\}$$

and the auxiliary constants :

- $w_i(\theta) = \frac{\hat{\rho}(\theta)}{2} w_i^{\text{Ref}}$ are the weights of the reference quadrature over the segment $[-1, 1]$.
- $W_{-1,n}(\theta) = \sum_{i=1}^n \frac{w_i(\theta)}{\rho_i(\theta)} = \sum_{i=1}^n \frac{w_i^{\text{Ref}}}{1+x_i^{\text{Ref}}}$.
- $W_{-2,n}(\theta) = \sum_{i=1}^n \frac{w_i(\theta)}{\rho_i^2(\theta)} = \frac{2}{\hat{\rho}(\theta)} \sum_{i=1}^n \frac{w_i^{\text{Ref}}}{(1+x_i^{\text{Ref}})^2}$.

Remark 4.2. In the case of Gauss-Legendre quadrature, the constants $W_{-1,n}$ and $W_{-2,n}$ can be computed analytically as:

$$W_{-1,n} = 2H_n, \quad W_{-2,n}(\theta) = \frac{2}{\hat{\rho}(\theta)} n(n+1), \quad P_n(\theta) = \frac{n(n+1)}{\hat{\rho}(\theta)} \quad (22)$$

With $H_n = \sum_{i=1}^n \frac{1}{i}$ the n -th harmonic number. These last two formulas are, for the moment, a conjecture. (Elaborate to prove it?)

Remark 4.3. Properties of the new weights $\tilde{w}_i(\theta)$

$$\text{For } n \geq 3, \quad \sum_{i=1}^n \tilde{w}_i(\theta) = \hat{\rho}(\theta) \quad (23)$$

This is easy to understand: either look at the definition of $\tilde{w}_i(\theta)$ and use the properties of $A_i(\theta)$ and $B_i(\theta)$, or integrate the constant function $F(\rho, \theta) = 1$ between 0 and $\hat{\rho}(\theta)$, which is equivalent. In a more general way, if the integrand F is not singular (i.e. $F \sim \mathcal{O}(1)$), then the quadrature rule is the standard quadrature as if reference quadrature was used to integrate the function F between 0 and $\hat{\rho}(\theta)$. If the integrand is singular as $1/\rho$, the customized quadrature just performs a Cauchy principal value. The true ingredients are the values of the Laurent coefficients F_{-1} and F_{-2} , which are computed numerically; they can be zero (up to the algorithm precision) or not, depending on the considered singularity.

5 Convergence and error study

5.1 Decomposition of the error

To begin our reasoning, assume that we have computed the Laurent's coefficients up to a relative error ϵ_{-k} , $k \in \{1, 2\}$:

$$\tilde{F}_{-k} = F_{-k}(1 + \epsilon_{-k}), \quad k \in \{1, 2\} \quad (24)$$

Now let's denote

$$R(\rho, \theta) = F(\rho, \theta) - \frac{F_{-1}(\theta)}{\rho} - \frac{F_{-2}(\theta)}{\rho^2} \quad (25)$$

$$\tilde{R}(\rho, \theta) = F(\rho, \theta) - \frac{\tilde{F}_{-1}(\theta)}{\rho} - \frac{\tilde{F}_{-2}(\theta)}{\rho^2} \quad (26)$$

$$(27)$$

$$I = \int_0^{2\pi} \int_0^{\hat{\rho}(\theta)} R(\rho, \theta) d\rho d\theta, \quad \tilde{I} = \int_0^{2\pi} \int_0^{\hat{\rho}(\theta)} \tilde{R}(\rho, \theta) d\rho d\theta \quad (28)$$

On one hand, after the theory of the Bernstein's ellipses, if we assume that R and $\tilde{R}$ are analytic, we can make a error majoration regarding the Gauss-Legendre quadrature of I and $\tilde{I}$:

$$|I - Q_n[R]| \leq \frac{64}{15} \frac{M}{(1 - \rho_B^{-2})\rho_B^{2n}} \quad (29)$$

$$|\tilde{I} - Q_n[\tilde{R}]| \leq \frac{64}{15} \frac{\tilde{M}}{(1 - \tilde{\rho}_B^{-2})\tilde{\rho}_B^{2n}} \quad (30)$$

On the other hand, we can subtract $\tilde{I}$ to I to get:

$$|I - \tilde{I}| \leq C(\hat{x}) \sum_{k=1}^2 \frac{\epsilon_{-k}}{\rho_{\min}^k} \|F_{-k}\|_{\infty} \quad (31)$$

where $C(\hat{x}) = \int_0^{2\pi} \hat{\rho}(\theta) d\theta$.

Thus,

$$|I - Q_n[\tilde{R}]| = |I - \tilde{I} + \tilde{I} - Q_n[\tilde{R}]| \leq |I - \tilde{I}| + |\tilde{I} - Q_n[\tilde{R}]| \leq C(\hat{x}) \sum_{k=1}^2 \frac{\epsilon_{-k}}{\rho_{\min}^k} \|F_{-k}\|_{\infty} + \frac{64}{15} \frac{\tilde{M}}{(1 - \tilde{\rho}_B^{-2})\tilde{\rho}_B^{2n}} \quad (32)$$

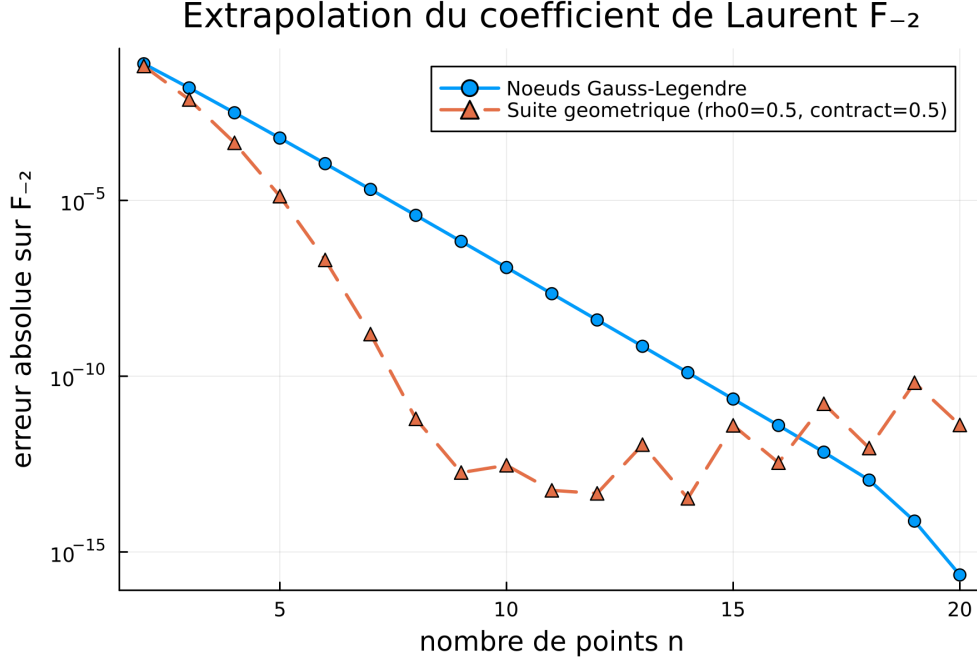


Figure 1: Convergence comparison between geometric and Gauss-Legendre sequences for the function $\rho \rightarrow \rho^2 \times 1/(\rho^2(1 + \rho))$ (realistic case)

5.2 Richardson extrapolation error

Lemma 5.1. *We can show that the following majorations hold for any sequence $\rho_i(\theta)$ of n radial points:*

$$|\tilde{F}_{-2} - F_{-2}| \leq \frac{\|\partial_\rho^n G\|_\infty}{n!} \prod_{i=1}^n \rho_i, \quad |\tilde{F}_{-1} - F_{-1}| \leq \left\{ \frac{\|\partial_\rho^{n+1} G\|_\infty}{(n+1)!} + P_n \frac{\|\partial_\rho^n G\|_\infty}{n!} \right\} \prod_{i=1}^n \rho_i \quad (33)$$

Which shows that the speed of convergence is conditionned by the product of the radial points $\prod_{i=1}^n \rho_i(\theta)$. In the case of a geometric sequence $\rho_i(\theta) = \hat{\rho}(\theta)q^{i-1}$, we have $\prod_{i=1}^n \rho_i(\theta) = \hat{\rho}^n(\theta)q^{\frac{n(n-1)}{2}}$, which is a very fast convergence. In the case of a sequence of Gauss-Legendre quadrature points, we have $\prod_{i=1}^n \rho_i(\theta) = \hat{\rho}^n(\theta) \frac{(n!)^2}{(2n)!}$, which is a slower convergence¹ (see Figures 1, 2, 3 and 4).

That's why geometric sequences can be preferred over Gauss-Legendre quadrature points for pure Richardson extrapolation.

Proof of lemma 5.1. Let $e(\rho) = G(\rho) - \Pi_{n-1}G(\rho)$ be the interpolation error of G by its Lagrange interpolant $\Pi_{n-1}G$. We have the following formula for the interpolation error, which is quite general on polynomial interpolation:

$$e(\rho) = G[\rho_1, \dots, \rho_n, \rho] \prod_{i=1}^n (\rho - \rho_i) \quad (34)$$

Where $G[\rho_1, \dots, \rho_n, \rho]$ is the divided difference of order n of G at the points $\rho_1, \dots, \rho_n, \rho$. By using the mean value theorem for divided differences, we can write

$$G[\rho_1, \dots, \rho_n, \rho] = \frac{G^{(n)}(\xi)}{n!} \quad (35)$$

For some $\xi \in [0, \hat{\rho}]$. Thus, we can write

¹We need to qualify this statement

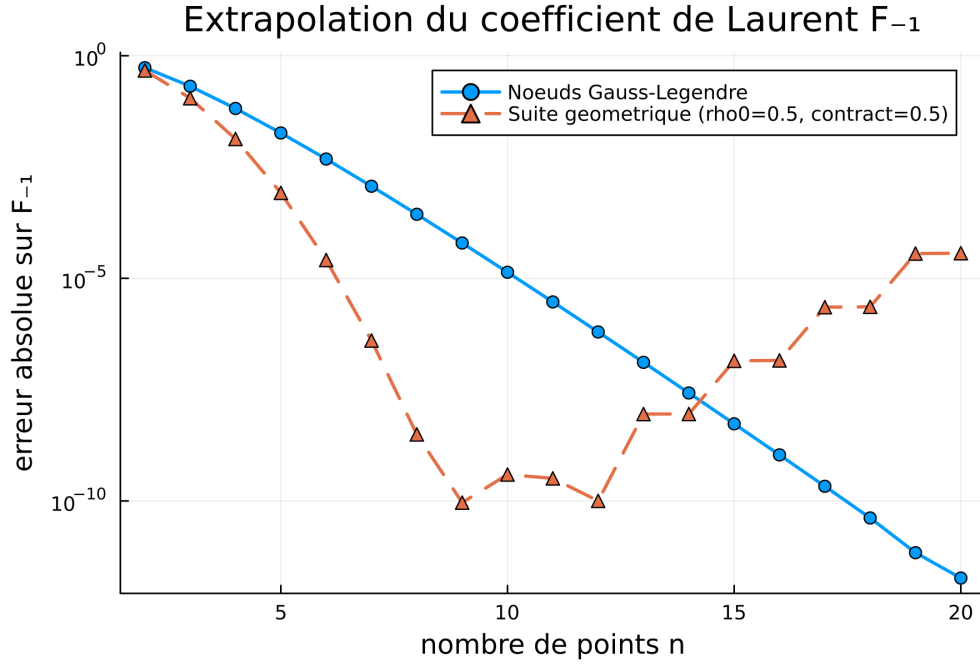


Figure 2: Convergence comparison between geometric and Gauss-Legendre sequences for the function $\rho \rightarrow \rho^2 \times 1/(\rho^2(1 + \rho))$ (realistic case)

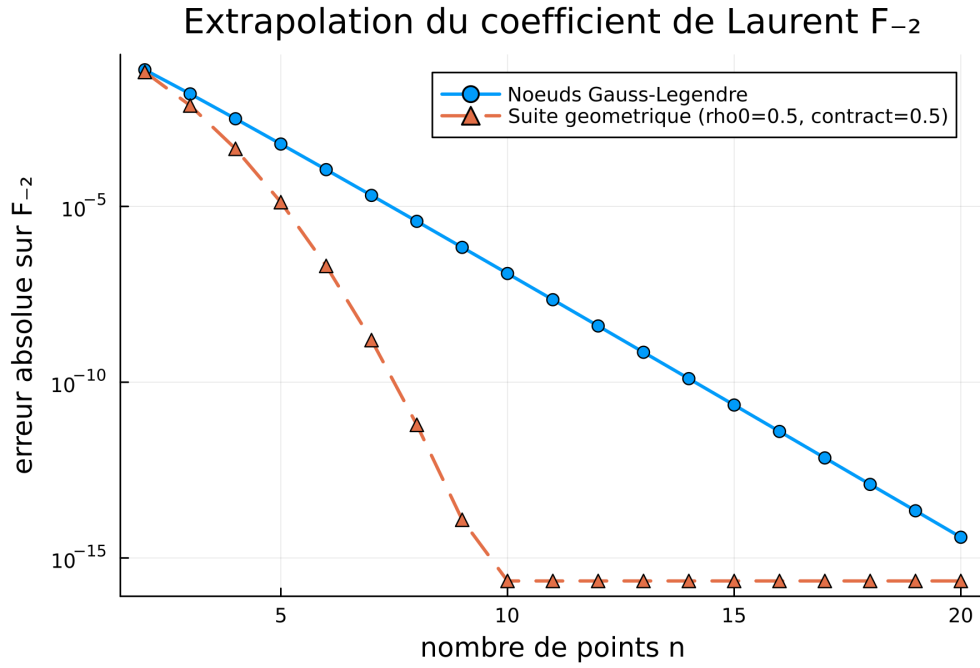


Figure 3: Convergence comparison between geometric and Gauss-Legendre sequences for the function $\rho \rightarrow 1/(1 + \rho)$ (ideal case)

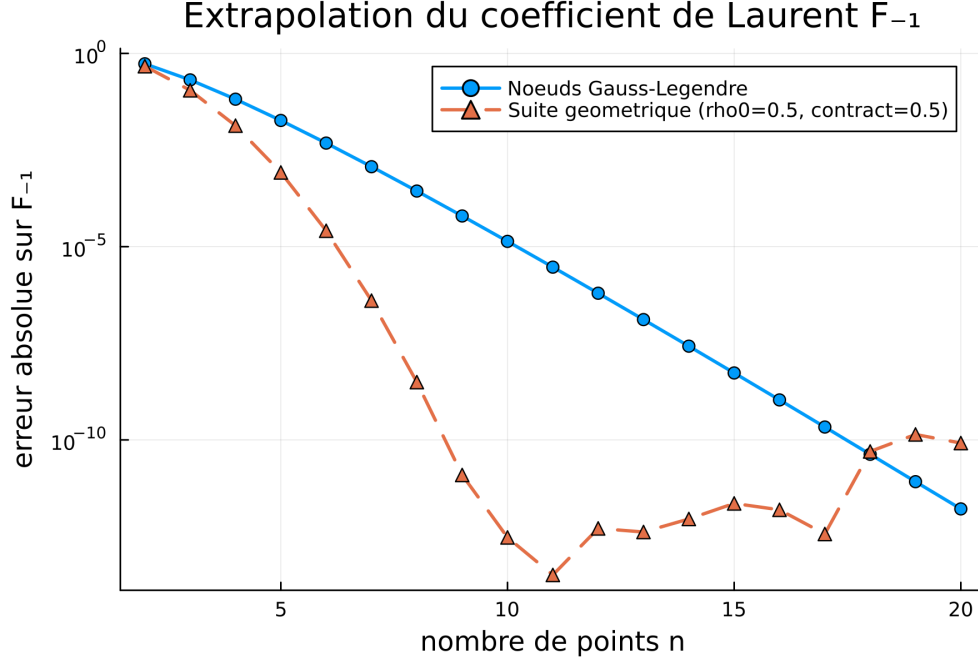


Figure 4: Convergence comparison between geometric and Gauss-Legendre sequences for the function $\rho \rightarrow 1/(1 + \rho)$ (ideal case)

$$\left| \tilde{F}_{-2} - F_{-2} \right| = |e(0)| \leq \frac{\|\partial_\rho^n G\|_\infty}{n!} \prod_{i=1}^n \rho_i \quad (36)$$

Then, taking the derivative of the interpolation error, we have

$$\begin{aligned} e'(\rho) &= \partial_\rho \left\{ G[\rho_1, \dots, \rho_n, \rho] \prod_{i=1}^n (\rho - \rho_i) \right\} \\ &= \partial_\rho \left\{ G[\rho_1, \dots, \rho_n, \rho] \right\} \prod_{i=1}^n (\rho - \rho_i) + G[\rho_1, \dots, \rho_n, \rho] \partial_\rho \left\{ \prod_{i=1}^n (\rho - \rho_i) \right\} \\ &= G[\rho_1, \dots, \rho_n, \rho] \prod_{i=1}^n (\rho - \rho_i) + G[\rho_1, \dots, \rho_n, \rho] P_n \prod_{i=1}^n (\rho - \rho_i) \\ \left| \tilde{F}_{-1} - F_{-1} \right| &= |e'(0)| \leq \left\{ \frac{\|\partial_\rho^{n+1} G\|_\infty}{(n+1)!} + P_n \frac{\|\partial_\rho^n G\|_\infty}{n!} \right\} \prod_{i=1}^n \rho_i \end{aligned}$$

using again the mean value theorem for $G[\rho_1, \dots, \rho_n, \rho]$ and $G[\rho_1, \dots, \rho_n, \rho, \rho]$.

□

5.3 Guiggiani's algorithm error, assuming Laurent's coefficients are exact

5.3.1 Gauss-Legendre quadrature general error estimate

To delve deeper into these general ideas, let's consider :

$$\tilde{I}_{\epsilon,h,n}(\theta) = Q_n[\tilde{R}_{\epsilon,h}](\theta) = \sum_{i=1}^n w_i(\theta) \left\{ F_h(\rho_i(\theta), \theta) - \frac{\tilde{F}_{-1,\epsilon,h}(\theta)}{\rho_i(\theta)} - \frac{\tilde{F}_{-2,\epsilon,h}(\theta)}{\rho_i^2(\theta)} \right\} \quad (37)$$

$$= \sum_{i=1}^n \left\{ F_h(\rho_i(\theta), \theta) - \frac{F_{-1,h}(\theta)}{\rho_i(\theta)} - \frac{F_{-2,h}(\theta)}{\rho_i^2(\theta)} \right\} w_i(\theta) - \sum_{k=1}^2 F_{-k,h}(\theta) \epsilon_{-k} \sum_{i=1}^n \frac{w_i(\theta)}{\rho_i^k(\theta)} \quad (38)$$

$$= Q_n[R_h](\theta) - \sum_{k=1}^2 F_{-k,h}(\theta) \epsilon_{-k} W_{-k,n}(\theta) \quad (39)$$

Where h is the mesh size. Thanks to standard results on the Gauss-Legendre quadrature for any function in $C^{2n}([0, \hat{\rho}(\theta)])$, we have the following error majoration (deleting the θ dependency for clarity):

$$Q_n[R_h] = \int_0^{\hat{\rho}} R_h(\rho) d\rho + \frac{2^{2n+1}(n!)^4}{(2n+1)[(2n)!]^3} R_h^{(2n)}(\rho_0) \quad (40)$$

for some $\rho_0 \in [0, \hat{\rho}]$. Our next goal is to show that the error term $R_h^{(2n)}(\rho_0)$ converges with the mesh size h and the number of quadrature points n . By using $G_h(\rho) = \rho^2 F_h(\rho) = \mathcal{O}_{\rho \rightarrow 0}(1)$, we can straightforwardly express R_h in terms of G , and bound its derivatives, assuming, as Guiggiani, that F_h is analytic in a neighborhood of the origin:

$$R_h(\rho) = \sum_{k=0}^{+\infty} F_k \rho^k, \quad \implies \quad R_h^{(2n)}(\rho) = \sum_{k=2n}^{+\infty} F_k \frac{k!}{(k-2n)!} \rho^{k-2n} \quad (41)$$

$$(42)$$

with $F_k = \frac{G_h^{(k+2)}(0)}{(k+2)!} \leq \frac{1}{(k+2)!} \|G_h^{(k+2)}\|_{\infty,[0,\hat{\rho}]}$ and $\rho^{k-2n} \leq \hat{\rho}^{k-2n}$, so that we can write

$$|R_h^{(2n)}(\rho)| \leq \sum_{k=2n}^{+\infty} k! \frac{\|G_h^{(k+2)}\|_{\infty,[0,\hat{\rho}]}}{(k+2)!(k-2n)!} \hat{\rho}^{k-2n} \quad (43)$$

5.3.2 Derivation of the regularized kernel G with respect to ρ

In this section, we will explicitly estimate $\partial^m G$ using complete chain rules derivatives in terms of the mesh size h , which corresponds to the size of the physical element τ in the physical space ; taking, for instance, the radius of the circumscribed circle of an arbitrary element τ of the mesh.

Definition 5.1. Let us first defined the following functions:

$$\begin{aligned} \phi_h : \hat{\tau} \subset \mathbb{R}^{d-1} &\rightarrow \mathbb{R} & \psi_{\hat{x}} : \mathbb{R} \times [0, 2\pi[&\rightarrow \mathbb{R}^{d-1} \\ \hat{x} &\mapsto \det M_h(\hat{x}) & (\rho, \theta) &\mapsto \hat{x} + \rho \mathbf{u}(\theta) \end{aligned}$$

$$\begin{aligned} \chi_h : \hat{\tau} \subset \mathbb{R}^{d-1} &\rightarrow \tau \subset \mathbb{R}^d \\ \hat{x} &\mapsto x \end{aligned}$$

where χ is the mapping from the reference space to the physical element, $\psi_{\hat{x}}$ is the mapping from polar coordinates to the reference space, and M_h is the metric tensor of the mapping χ_h , defined as:

$$M_h(\hat{x}) = D\chi_h(\hat{x})^T D\chi_h(\hat{x}), \quad (M_h)_{ab} = \sum_{k=1}^d \frac{\partial \chi_{h,k}}{\partial \hat{x}_a} \frac{\partial \chi_{h,k}}{\partial \hat{x}_b}, \quad (a, b) \in \{1, \dots, d-1\}^2 = \{1, 2\}^2 \quad (44)$$

Finally, the vector $\mathbf{u}(\theta)$ is the unit trigonometric vector defined as $\mathbf{u}(\theta) = (\cos \theta, \sin \theta)$ for our case $d = 3$.

Lemma 5.2. *Using the Faà di Bruno formula for functions of the form $h(x) = f[g_1(x), \dots, g_m(x)]$ (see [constantine_multivariate_1996]), one can write, for any function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$,*

$$\partial_\rho^k (f \circ \psi_{\hat{x}})(\rho, \theta) = \sum_{i=0}^k \binom{k}{i} \cos^i(\theta) \sin^{k-i}(\theta) \frac{\partial^k f}{\partial \hat{x}_1^i \partial \hat{x}_2^{k-i}}(\psi_{\hat{x}}(\rho, \theta)) \quad (45)$$

We can then expand G as

$$G(\rho, \theta) = H_h(\rho, \theta) J_h \circ \psi_{\hat{x}}(\rho, \theta) \quad (46)$$

where

$$H_h(\rho, \theta) = \rho^3 K(x = \chi_h(\hat{x}), y = \chi_h \circ \psi_{\hat{x}}(\rho, \theta)) \hat{u} \circ \psi_{\hat{x}}(\rho, \theta), \quad J_h(\hat{x}) = \sqrt{\phi_h(\hat{x})} = \sqrt{\det M_h(\hat{x})} \quad (47)$$

Lemma 5.3. *We have the following useful factorization:*

$$\mathbf{r} = \rho \mathbf{A}_{\hat{x}}(\rho, \theta), \quad \mathbf{A}_{\hat{x}}(\rho, \theta) = \sum_{m \geq 1} \frac{\rho^{m-1}}{m!} D^m \chi_h(\hat{x}) :_m (\mathbf{u}(\theta)^{m \otimes}) \quad (48)$$

Homogeneous kernels. When the kernel is homogeneous, i.e

$$\begin{aligned} \exists \hat{K}, K(x, y) &= \frac{1}{r^s} \hat{K}(\hat{\mathbf{r}}, n_x, n_y), \quad \forall n_x, n_y \in \mathbb{S}^d, z \rightarrow \hat{K}(z, n_x, n_y) \text{ is of class } \mathcal{C}^\infty(\mathbb{R}^d) \\ \text{i.e } \forall \alpha \in \mathbb{N}^d, \quad \exists C_K(|\alpha|) > 0, \quad \left| \frac{\partial^{|\alpha|} \hat{K}}{\partial z^\alpha}(z, n_x, n_y) \right| &\leq C_K(|\alpha|), \quad \forall z \in \mathbb{R}^d \end{aligned} \quad (49)$$

Then we have

$$\rho^s K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho, \theta)) = \left(\frac{\rho}{r}\right)^s \hat{K}(\hat{\mathbf{A}}_{\hat{x}}(\rho, \theta), n \circ \chi_h(\hat{x}), n \circ \chi_h \circ \psi_{\hat{x}}(\rho, \theta)) \quad (50)$$

$$= \frac{1}{\|\mathbf{A}_{\hat{x}}(\rho, \theta)\|^s} \hat{K}(\hat{\mathbf{A}}_{\hat{x}}(\rho, \theta), n \circ \chi_h(\hat{x}), n \circ \chi_h \circ \psi_{\hat{x}}(\rho, \theta)) \quad (51)$$

$$= A_{\hat{x}}(\rho, \theta)^{-s} \hat{K}(\hat{\mathbf{A}}_{\hat{x}}(\rho, \theta), n \circ \chi_h(\hat{x}), n \circ \chi_h \circ \psi_{\hat{x}}(\rho, \theta)) \quad (52)$$

Where s is the order of singularity of the function $\rho \rightarrow K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho, \theta))$ at the origin, such that $K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho, \theta)) = \mathcal{O}_{\rho \rightarrow 0}(\rho^{-s})$. The idea was originally used for hypersingular 3D kernels, where $s = 3$ (see Guiggiani, 1992), but we write the computations with general order s because the idea remains the same.

We will denote $\hat{K}(\hat{\mathbf{A}}_{\hat{x}}(\rho, \theta), n \circ \chi_h(\hat{x}), n \circ \chi_h \circ \psi_{\hat{x}}(\rho, \theta))$ as $\hat{K}_{\hat{x}}(\rho, \theta)$ for clarity, keeping in mind that this is the kernel evaluated in polar coordinates in parameter space centered at point $\hat{x}$.

As the kernel K is regularized by the term ρ^3 in H_h , we can ensure that $H_h = \mathcal{O}_{\rho \rightarrow 0}(1)$. Applying the Leibniz formula, we have

$$\partial_\rho^m G(\rho, \theta) = \sum_{k=0}^m \binom{m}{k} \partial_\rho^k H_h(\rho, \theta) \partial_\rho^{m-k} (J_h \circ \psi_{\hat{x}})(\rho, \theta) \quad (53)$$

With:

$$\begin{aligned}\partial_\rho^k H_h(\rho, \theta) &= \sum_{i=0}^k \binom{k}{i} \partial_\rho^i \left(A_{\hat{x}}(\rho, \theta)^{-s} \hat{K}_{\hat{x}}(\rho, \theta) \right) \partial_\rho^{k-i} (\hat{u} \circ \psi_{\hat{x}})(\rho, \theta) \\ \partial_\rho^{m-k} (J_h \circ \psi_{\hat{x}})(\rho, \theta) &\leq \sum_{j=0}^{m-k} \binom{m-k}{j} \left| \frac{\partial^{m-k} J_h}{\partial \hat{x}_1^j \partial \hat{x}_2^{m-k-j}} (\psi_{\hat{x}}(\rho, \theta)) \right|\end{aligned}$$

To bound the $(m-k)$ -th order polar derivatives of J_h , we used the lemma 5.2.

BOUNDING OF THE CARTESIAN DERIVATIVES OF J_h : to ensure non-degeneracy of the elements, we can assume that

$$\exists C_Q > 0, \quad \forall h > 0, \quad \forall m > 0, \quad \forall \hat{x} \in \hat{\tau}, \quad \|D^m \chi_h(\hat{x})\|_F \leq C_Q h^m \quad (54)$$

Where $D^m \chi(\hat{x})$ is the tensor of order m containing all the m -th order derivatives of χ_h at $\hat{x}$, defined as

$$(D^m \chi_h(\hat{x}))_{i, \alpha_1, \dots, \alpha_m} = \frac{\partial^m \chi_{h,i}}{\partial \hat{x}_{\alpha_1} \dots \partial \hat{x}_{\alpha_m}}(\hat{x}), \quad i \in \{1, 2, 3\}, \quad (\alpha_1, \dots, \alpha_m) \in \{1, 2\}^m \quad (55)$$

Then, we recall the shape-regularity condition which is the upper bound equivalent of the non-degeneracy condition 54:

$$\exists \widetilde{C_Q} > 0, \quad \forall h > 0, \quad \forall (\hat{x}, \hat{y}) \in \hat{\tau}^2, \quad \|\chi_h(\hat{x}) - \chi_h(\hat{y})\| \geq \widetilde{C_Q} h \|\hat{x} - \hat{y}\|. \quad (56)$$

Lemma 5.4.

$$|\partial^\nu \phi_h(\hat{x})| \leq (2C_Q^2 d)^{d-1} (d-1)^{|\nu|} h^{|\nu|+2(d-1)} (d-1)!, \quad \forall \nu \in \mathbb{N}^{d-1}, \forall \hat{x} \in \hat{\tau} \quad (57)$$

Proof of lemma 5.4. By the Leibniz formula for the determinant, we have

$$\phi_h = \det M_h = \sum_{\sigma \in \mathfrak{S}_{d-1}} \text{sgn}(\sigma) \prod_{i=1}^{d-1} (M_h)_{i, \sigma(i)} \quad (58)$$

Furthermore, by the generalized Leibniz formula for the derivatives of a product, we have

$$\partial_{\hat{x}_j}^k \prod_{i=1}^{d-1} (M_h)_{\sigma(i), i} = \sum_{\substack{k_1, \dots, k_{d-1} \in \mathbb{N} \\ k_1 + \dots + k_{d-1} = k}} \frac{k!}{k_1! \dots k_{d-1}!} \prod_{i=1}^{d-1} \partial_{\hat{x}_j}^{k_i} (M_h)_{\sigma(i), i} \quad (59)$$

Where each term $\partial_{\hat{x}_j}^{k_i} (M_h)_{\sigma(i), i}$ can be bounded by the non-degeneracy condition 54:

$$\partial_{\hat{x}_j}^{k_i} (M_h)_{\sigma(i), i} = \sum_{\ell=1}^d \left(\frac{\partial^{k_i+1} \chi_{h, \ell}}{\partial \hat{x}_{\sigma(i)} \partial \hat{x}_j^\ell} \frac{\partial \chi_{h, k}}{\partial \hat{x}_i} + \frac{\partial \chi_{h, k}}{\partial \hat{x}_{\sigma(i)}} \frac{\partial^{k_i+1} \chi_{h, \ell}}{\partial \hat{x}_i \partial \hat{x}_j} \right) \leq 2C_Q^2 d h^{k_i+2} \quad (60)$$

Thus, we can write

$$|\partial_{\hat{x}_j}^k \prod_{i=1}^{d-1} (M_h)_{\sigma(i), i}| \leq (2C_Q^2 d)^{d-1} h^{k+2(d-1)} \sum_{\substack{k_1, \dots, k_{d-1} \in \mathbb{N} \\ k_1 + \dots + k_{d-1} = k}} \frac{k!}{k_1! \dots k_{d-1}!} \quad (61)$$

$$= (2C_Q^2 d)^{d-1} h^{k+2(d-1)} (d-1)^k \quad (62)$$

Since $\#\mathfrak{S}_{d-1} = (d-1)!$, we can conclude that

$$|\partial_{\hat{x}_j}^k \phi_h| \leq (2C_Q^2 d)^{d-1} (d-1)^k h^{k+2(d-1)} (d-1)! \quad (63)$$

Which leads to the desired result. $\square$

Lemma 5.5.

$$|\partial^\nu J_h(\hat{x})| \leq h^{d+|\nu|-1} \mathcal{J}_\nu, \quad \forall \nu \in \mathbb{N}^{d-1}, \forall \hat{x} \in \hat{\tau} \quad (64)$$

Where $\mathcal{J}_\nu = \sum_{r=1}^n \sqrt{2r!} \nu! \widetilde{C_Q}^{(d-1)(1-2r)} (2C_Q^2 d)^d (d-1)^n (d-1)! P_{\nu, r}$.

Proof of lemma 5.5. By the generalized Faà di Bruno formula, and the lemma 5.4, we have

$$|\partial^\nu J_h| \leq \sum_{r=1}^n \sqrt{2}r! (\widetilde{c_Q}h)^{(d-1)(1-2r)} \nu! (2C_Q^2d)^r (d-1)^n (d-1)! h^{n+2(d-1)} P_{\nu,r} \quad (65)$$

$$= h^{d+|\nu|-1} \sum_{r=1}^n \sqrt{2}r! \nu! \widetilde{c_Q}^{(d-1)(1-2r)} (2C_Q^2d)^d (d-1)^n (d-1)! P_{\nu,r} \quad (66)$$

$$(67)$$

as desired. $\square$

Corollary 5.1. By a simple application of the lemma 5.5 and the lemma 5.2, we can write

$$|\partial_\rho^n (J_h \circ \psi_{\hat{x}})(\rho, \theta)| \leq h^{d+n-1} \mathcal{J}_n, \quad \forall n \in \mathbb{N}, \quad (68)$$

$$\text{Where } \mathcal{J}_n = \sum_{k=0}^n \binom{n}{k} \mathcal{J}_{(k, n-k)}.$$

Lemma 5.6.

$$\forall \lambda \in \mathbb{N}^d, \forall x \in \mathbb{R}^d \setminus \{O\}, \quad |\partial^\lambda \|x\|_2| \leq \lambda! \left(1 + \frac{1}{\sqrt{2}}\right) \sqrt{2}^{|\lambda|} \|x\|_2^{1-|\lambda|} \quad (69)$$

Where $\lambda! = \prod_{i=1}^d \lambda_i!$ and $|\lambda| = \sum_{i=1}^d \lambda_i$.

Remark 5.1. The majoration constant $C_\lambda = \lambda! \left(1 + \frac{1}{\sqrt{2}}\right) \sqrt{2}^{|\lambda|}$ is not optimal, but it is better than the one that would be obtained after using the Faà di Bruno formula.

Proof of lemma 5.6. Let $x \in \mathbb{R}^d \setminus \{O\}$. First, the function $\|\cdot\|$ is analytic at x , so by extension it admits an holomorphic continuation in a neighborhood of x in $\mathbb{C}^d \setminus \Sigma$, where Σ is the singular complex variety defined as $\Sigma = \{z \in \mathbb{C}^d, \sum_{i=1}^d z_i^2 = 0\}$. By the Cauchy's estimate, if we have an open ball $B(x, R) \subset \mathbb{C}^d \setminus \Sigma$, we have the following majoration for any multi-index $\lambda \in \mathbb{N}^d$:

$$|\partial^\lambda \|x\|_2| \leq \frac{\lambda!}{R^{|\lambda|}} \sup_{z \in B(x, R)} \|z\|_2 \quad (70)$$

If we want to make this inequality more explicit, we should find such a radius R that depends only on x . Let's compute the distance from x to the singular variety Σ . Let $z = y + ic \in \Sigma$.

$$\begin{aligned} z \in \Sigma &\iff z \cdot z = 0 \\ &\iff (y + ic) \cdot (y + ic) = 0 \\ &\iff |y|^2 - |c|^2 = 2i y \cdot c \\ &\iff \begin{cases} |y|^2 - |c|^2 = 0, \\ y \cdot c = 0. \end{cases} \end{aligned}$$

Furthermore, we have

$$\begin{aligned} |x - z|^2 &= |x - y - ic|^2 \\ &= |x - y|^2 + |c|^2 \end{aligned}$$

So $\text{dist}(x, \Sigma)^2 = \inf_{\substack{y, c \in \mathbb{R}^d \\ |y|=|c|, y \perp c}} |x - y|^2 + |c|^2 = \inf_{\substack{y, c \in \mathbb{R}^d \\ y \perp c}} |x - y|^2 + |y|^2$. For $d \geq 2$, for every $y \in \mathbb{R}^d$ it always exists a $c \in \mathbb{R}^d$ such that $y \perp c$. So we can write

$$\text{dist}(\mathbf{x}, \Sigma)^2 = \inf_{\mathbf{y} \in \mathbb{R}^d} |\mathbf{x} - \mathbf{y}|^2 + |\mathbf{y}|^2$$

If we let $f(\mathbf{y}) = |\mathbf{x} - \mathbf{y}|^2 + |\mathbf{y}|^2$, we have $\nabla f(\mathbf{y}) = 4\mathbf{y} - 2\mathbf{x}$. The minimum is reached for $\nabla f(\mathbf{y}) = 0$, i.e. $\mathbf{y} = \frac{\mathbf{x}}{2}$. Thus, we have

$$\boxed{\text{dist}(\mathbf{x}, \Sigma) = \frac{|\mathbf{x}|}{\sqrt{2}}}$$

This formula gives us how close we are from the "danger zone" in the complex space. Thus, we let $R = \theta \|\mathbf{x}\|_2$ with $0 < \theta < 1/\sqrt{2}$ as a relaxing parameter, as well as $C(\theta) = \frac{\lambda!}{R^{|\lambda|}} \sup_{\mathbf{z} \in B(\mathbf{x}, R)} \|\mathbf{z}\|_2$. For all $\mathbf{z} \in B(\mathbf{x}, R)$, we have

$\|\mathbf{z}\|_2 \leq \|\mathbf{x}\|_2 + R = (1 + \theta) \|\mathbf{x}\|_2^2$, so that we can write

$$\sup_{\mathbf{z} \in B(\mathbf{x}, R)} \|\mathbf{z}\|_2 \leq (1 + \theta) \|\mathbf{x}\|_2$$

and also $C(\theta) \leq \lambda! \frac{1 + \theta}{\theta^{|\lambda|}} \|\mathbf{x}\|_2^{1-|\lambda|} = \lambda! g(\theta) \|\mathbf{x}\|_2^{1-|\lambda|}$. The function g is decreasing on $]0, 1/\sqrt{2}[$, so the best majorant that we can take is the minimum majorant reached at the maximum value of θ , i.e. $\theta = 1/\sqrt{2}$, which gives us $g(1/\sqrt{2}) = (1 + 1/\sqrt{2})\sqrt{2}^{|\lambda|}$. Thus, we have the desired majoration [70](#). □

Corollary 5.2. *Let $\mathbf{f} : \mathbb{R} \rightarrow \mathbb{R}^d$ be a function of class $\mathcal{C}^{|\lambda|}$, with $\lambda \in \mathbb{N}^d$. Then, the following bound holds, which is a consequence of the three following statements:*

- if $\mathbf{v}, \mathbf{u} \in \mathbb{R}^d$, then vectorial exponentiation can be simply bounded by $|\mathbf{u}^{\mathbf{v}}| \leq \|\mathbf{u}\|_2^{\|\mathbf{v}\|_1} = \|\mathbf{u}\|_2^{|\mathbf{v}|}$
- the Faà di Bruno formula for the composition of functions $h = g \circ f$ with $g : \mathbb{R}^d \rightarrow \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}^d$
- the previous lemma [5.6](#)

$$|\partial^n \mathbf{f}(\rho)|_2 \leq \sum_{1 \leq |\lambda| \leq n} \lambda! C_\lambda \|f(\rho)\|_2^{1-|\lambda|} \sum_{p(n, \lambda)} n! \prod_{j=1}^n \frac{\|\partial^j \mathbf{f}(\rho)\|_2^{|\mathbf{k}_j|}}{\mathbf{k}_j! j^{|\mathbf{k}_j|}} \quad (71)$$

Lemma 5.7. *We can achieve the following upper bound for the derivatives of scalar $A_{\hat{x}}(\rho, \theta)$:*

$$|\partial_\rho^n A_{\hat{x}}(\rho, \theta)| \leq h^{n+1} \mathcal{C}_n \quad (72)$$

with $\mathcal{C}_n = \mathcal{C}_n(h, \theta, \hat{x}, \chi_h) = \sum_{1 \leq |\lambda| \leq n} C_\lambda |p(n, \lambda)| e^{|\lambda| h \hat{\rho}(\theta)} P_{n, \lambda} \prod_{i=1}^n (i+1)^{-|\mathbf{k}_i|} \widetilde{c_Q} \left(\frac{C_Q}{\widetilde{c_Q}} \right)^{|\lambda|}$ and $C_k = k! \left(1 + \frac{1}{\sqrt{2}}\right) \sqrt{2}^k$.

Also, we have $\mathcal{C}_n = \mathcal{O}_{h \rightarrow 0}(1)$ and $\mathcal{C}_n \neq \mathcal{O}_{h \rightarrow 0}(h)$.

BOUNDING OF THE DERIVATIVES OF THE REGULARIZED KERNEL: Let's differentiate $A_{\hat{x}}(\rho, \theta)^{-s} \hat{K}_{\hat{x}}(\rho, \theta)$. Using both the Leibniz and the Faà di Bruno formulas, we can write

$$\left(\frac{f}{g^s} \right)^{(n)} = \sum_{i=0}^n \binom{n}{i} f^{(n-i)} \sum_{k=0}^i (-1)^k g^{-s-k} s^{(k)} \sum_{p(i, k)} \frac{i!}{\prod_{j=1}^i k_j!} \prod_{j=1}^i \left(\frac{\partial^j g}{j!} \right)^{k_j} \quad (73)$$

With, by convention,

²Here, we perform a non optimal majoration.

$$\sum_{p(i,0)} \frac{i!}{\prod_{j=1}^i k_j!} \prod_{j=1}^i \left(\frac{\partial^j g}{j!} \right)^{k_j} = \delta_{i,0},$$

so that we can retrieve, for $n = 0$, the original function $\frac{f}{g^s}$, and where $s^{(k)} = s(s+1) \cdots (s+k-1) = \frac{(s+k-1)!}{(s-1)!}$ is the rising factorial.

Lemma 5.8. *We have another upper bounds:*

$$\left\| \partial_\rho^\ell \mathbf{A}_{\hat{x}}(\rho, \theta) \right\|_2 \leq C_Q \frac{h^{\ell+1}}{\ell+1} e^{\hat{\rho}(\theta)h}, \quad \left\| \partial_\rho^j \hat{\mathbf{A}}_{\hat{x}}(\rho, \theta) \right\|_2 \leq h^j e^{\hat{\rho}(\theta)h} D_j \quad (74)$$

$$\text{With } D_j = \sum_{0 \leq k \leq i \leq j} \frac{(j)_i k!}{j-i+1} \frac{C_Q}{\widetilde{c_Q}^{1+k}} c_{i,k} P_{i,k}$$

For the next sections, let's forget about the two normal vectors $n \circ \chi_h(\hat{x}), n \circ \chi_h \circ \psi_{\hat{x}}(\rho, \theta) \in \mathbb{S}^d$, and consider the kernel $\hat{K}$ as a function of only one variable $z \in \mathbb{R}^d$. This is possible because the two normal vectors are bounded by construction of the mesh, and we can include them in the constant $C_K(\lambda)$. Thus, we can write $\hat{K}(\hat{\mathbf{A}}_{\hat{x}}(\rho, \theta), n \circ \chi_h(\hat{x}), n \circ \chi_h \circ \psi_{\hat{x}}(\rho, \theta)) = \hat{K}_{\hat{x}}(\rho, \theta) = \hat{K} \circ \hat{\mathbf{A}}_{\hat{x}}(\rho, \theta)$.

Corollary 5.3. *This follows:*

$$\left\| \partial_\rho^m (\hat{K} \circ \hat{\mathbf{A}}_{\hat{x}})(\rho) \right\| \leq h^m \mathcal{L}_m(\theta, h) \quad (75)$$

With $\mathcal{L}(\theta, h) = \underset{h \rightarrow 0}{O}(1)$ and $\mathcal{L}(\theta, h) \neq \underset{h \rightarrow 0}{O}(h)$.

More explicitly, we have

$$\mathcal{L}_m(\theta, h) = m! \sum_{1 \leq |\lambda| \leq m} C_K(\lambda) e^{\hat{\rho}(\theta)h|\lambda|} P_{m,\lambda} \sum_{p(m,\lambda)} \prod_{j=1}^m D_j^{|\mathbf{k}_j|} \quad (76)$$

Before diving into the summations and product in 73, let's establish the following straightforward bound, using the mesh regularity condition 56:

$$A_{\hat{x}}(\rho, \theta)^{-s-k} = (r/\rho)^{-s-k} \leq (c_Q h)^{-s-k} \quad (77)$$

Now, our next goal is to bound the term $P_{p(i,k)} = \frac{1}{\prod_{j=1}^i k_j!} \prod_{j=1}^i \left(\frac{\partial_\rho^j A_{\hat{x}}(\rho, \theta)}{j!} \right)^{k_j}$.

$$P_{p(i,k)} = \prod_{j=1}^i \frac{(\partial_\rho^j A_{\hat{x}}(\rho, \theta))^{k_j}}{j!^{k_j} k_j!} \leq \prod_{j=1}^i \frac{(h^{j+1} \mathcal{C}_j)^{k_j}}{j!^{k_j} k_j!} \leq \prod_{j=1}^i (h^{j+1} \mathcal{C}_j)^{k_j} P_{i,k} \quad (78)$$

With $\prod_{j=1}^i (h^{j+1} \mathcal{C}_j)^{k_j} = h^{k+i} \prod_{j=1}^i \mathcal{C}_j^{k_j}$, by definition of the set $p(i, k)$.

Thus,

$$\sum_{p(i,k)} P_{p(i,k)} \leq h^{k+i} \mathcal{C}_{i,k} P_{i,k}, \quad \mathcal{C}_{i,k} = \sum_{p(i,k)} \prod_{j=1}^i \mathcal{C}_j^{k_j}, \quad k \stackrel{\text{recall}}{=} \sum_{j=1}^i k_j \quad (79)$$

Finally,

$$\partial_\rho^j \left(A_{\hat{x}}(\rho, \theta)^{-s} \hat{K}_{\hat{x}}(\rho, \theta) \right) \leq h^{j-s} c_Q^{-s} \sum_{0 \leq k \leq i \leq j} (j)_i s^{(k)} \mathcal{L}_{j-i}(\theta, h) \mathcal{C}_{i,k} P_{i,k} \quad (80)$$

$$= h^{j-s} \mathcal{K}_{j,s}(\theta, h) \quad (81)$$

With $(k)_i = \frac{k!}{(k-i)!}$ the falling factorial.

BOUNDING OF THE DERIVATIVES OF $\hat{u} \circ \psi_{\hat{x}}$: Using the lemma 5.2, we can write

$$|\partial_\rho^{k-i}(\hat{u} \circ \psi_{\hat{x}})(\rho, \theta)| \leq \sum_{j=0}^{k-i} \binom{k-i}{j} \frac{\partial^{k-i} \hat{u}}{\partial \hat{x}_1^j \partial \hat{x}_2^{k-i-j}}(\psi_{\hat{x}}(\rho, \theta)) \quad (82)$$

Lemma 5.9. Let x_τ be a fixed point inside τ , such that $\forall h > 0, \tau_h \subset B_h(x_\tau)$. If u is $\mathcal{C}^m$ in a vicinity of x_τ , then

$$\forall \lambda \in \mathbb{N}^d, |\lambda| \leq m, \quad \|\partial^\lambda u\|_{\infty, \tau_h} := u_\lambda = O_{h \rightarrow 0}(1) \quad (83)$$

Lemma 5.10. Let $\nu_{ij} = (i, j-i)$. Then we have

$$\|\partial^{\nu_{ij}} \hat{u}(\hat{y})\| \leq h^j \hat{u}_{ij} \quad (84)$$

With $\hat{u}_{ij} = i! (j-i)! \sum_{1 \leq |\lambda| \leq j} u_\lambda C_Q^{|\lambda|} \sum_{s=1}^j P_{\nu_{ij}, \lambda}^s$

Lemma 5.11.

$$\|\partial_\rho^j(\hat{u}_h \circ \psi_{\hat{x}})(\rho, \theta)\| \leq h^j \mathcal{U}_j \quad (85)$$

With $\mathcal{U}_j = j! \sum_{i=0}^j \sum_{1 \leq |\lambda| \leq j} \sum_{s=1}^j u_\lambda C_Q^{|\lambda|} P_{\nu_{ij}, \lambda}^s$

$$\begin{aligned} \|\partial_\rho^k H_h(\rho, \theta)\| &\leq \sum_{i=0}^k \binom{k}{i} h^{i-s} \mathcal{K}_{i,s}(\theta, h) \|\partial_\rho^{k-i}(\hat{u} \circ \psi_{\hat{x}})(\rho, \theta)\| \\ &\leq \sum_{i=0}^k \binom{k}{i} h^{i-s} \mathcal{K}_{i,s}(\theta, h) h^{k-i} \mathcal{U}_{k-i} \\ &= h^{k-s} \sum_{i=0}^k \binom{k}{i} \mathcal{K}_{i,s}(\theta, h) h^{k-i} \mathcal{U}_{k-i} \\ &= h^{k-s} \mathcal{H}_{k,s}(\theta, h) \end{aligned}$$

Finally, the desired bound on the derivatives of the G function is given by:

$$\boxed{\|\partial_\rho^m G_h(\rho, \theta)\| \leq h^{m+d-s-1} \mathcal{G}_{m,s}(\theta, h)} \quad (86)$$

With $\mathcal{G}_{m,s}(\theta, h) = \sum_{k=0}^m \binom{m}{k} \mathcal{H}_{k,s}(\theta, h) \mathcal{J}_{m-k}$. This bound is optimal with h , i.e. $\mathcal{G}_{m,s}(\theta, h) \begin{cases} = O_{h \rightarrow 0}(1) \\ \neq O_{h \rightarrow 0}(h) \end{cases}$. Let's use another notation, by noticing that the constant $\mathcal{G}_{m,s}(\theta, h)$ can be made independant of h :

$$\|\partial_\rho^m G_h(\rho, \theta)\| \leq h^{m+d-s-1} \mathcal{G}_{m,s}(\theta) \lesssim h^{m+d-s-1} \quad (87)$$

Corollary 5.4.

$$F_k \lesssim h^{k+d-s+1} \quad (88)$$

Corollary 5.5.

$$\partial_\rho^{2n} R_h \lesssim h^{2n+d-s+1} \quad (89)$$

Thus, we have this important claim:

$$Q_n[R_h] = \int_0^{\hat{\rho}} R_h(\rho) d\rho + \underset{h \rightarrow 0}{O}(h^{2n+d-s+1}) \quad (90)$$

5.3.3 Numerical validation

Let us consider the fixed smooth surface $\mathcal{S} = \{(x_1, x_2, \varphi(x_1, x_2)), (x_1, x_2) \in \mathbb{R}^2\}$, where

$$\varphi(x_1, x_2) = 0.4x_1^2 - 0.3x_1x_2 + 0.25x_2^2 + 0.2x_1^3 - 0.1x_2^3, \quad (91)$$

as shown in figure 6. We then consider the family of elements $\tau_h = \chi_h(\hat{\tau})$, $\hat{\tau} = [0, 1]^2$ whose curvatures doesn't depend on the size h , defined by the mapping

$$\chi_h(\hat{\mathbf{y}}) = (y_1, y_2, \varphi(y_1, y_2)), \quad (y_1, y_2) = h(\hat{\mathbf{y}} - \hat{\mathbf{c}}), \quad \hat{\mathbf{c}} = \left(\frac{1}{2}, \frac{1}{2}\right), \quad (92)$$

so that τ_h is the portion of $\mathcal{S}$ lying above the square $[-h/2, h/2]^2$. Its diameter is $h\sqrt{2}$ and it shrinks, as $h \rightarrow 0$, onto the fixed point $\mathbf{x}_\tau = (0, 0, 0)$ of $\mathcal{S}$. The essential point is that φ is evaluated at the *physical* coordinates $(y_1, y_2) = h(\hat{\mathbf{y}} - \hat{\mathbf{c}})$ and not at the reference ones: the curvature of τ_h at $\mathbf{x}_\tau$ is therefore the same for every h , and the family is mesh-regular in the sense of section 5.3.2. This family of elements, as well as the free surface $\mathcal{S}$, is shown in figure 6.

The associated geometric quantities read

$$D\chi_h(\hat{\mathbf{y}}) = h \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \partial_1\varphi(y_1, y_2) & \partial_2\varphi(y_1, y_2) \end{pmatrix}, \quad J_h(\hat{\mathbf{y}}) = h^2 \sqrt{1 + \|\nabla\varphi(y_1, y_2)\|^2}, \quad \mathbf{n}(\hat{\mathbf{y}}) = \frac{(-\partial_1\varphi, -\partial_2\varphi, 1)}{\sqrt{1 + \|\nabla\varphi\|^2}}, \quad (93)$$

which exhibit the two scalings used throughout section 5.3.2: $\|D\chi_h\| = \mathcal{O}(h)$ and $J_h = \mathcal{O}(h^{d-1})$.

Finally, the density is taken to be the restriction to τ_h of a *fixed physical* scalar function u , so that $\hat{u}_h = u \circ \chi_h$:

$$u(\mathbf{x}) = 1 + 0.7x_1 - 0.4x_2 + 0.9x_3. \quad (94)$$

The singular kernel is taken to be the classical Laplace hypersingular 3D kernel. Also, we fixed the angle value at $\theta = 0.7$ rad and the source point $\hat{\mathbf{x}} = (0.37, 0.53)$ in the reference element $\hat{\tau}$, which are just random values.

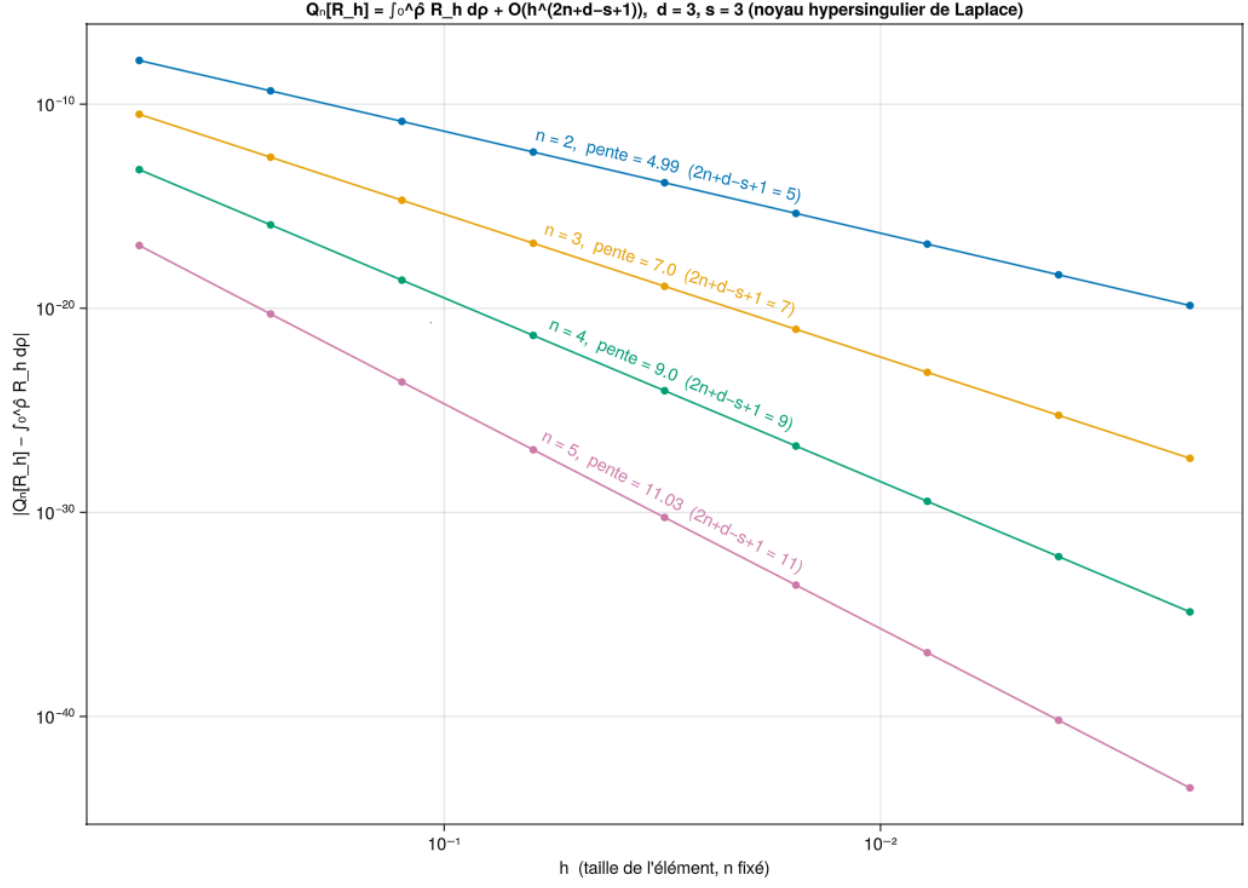


Figure 5: Convergence rate of the quadrature error for different values of n .

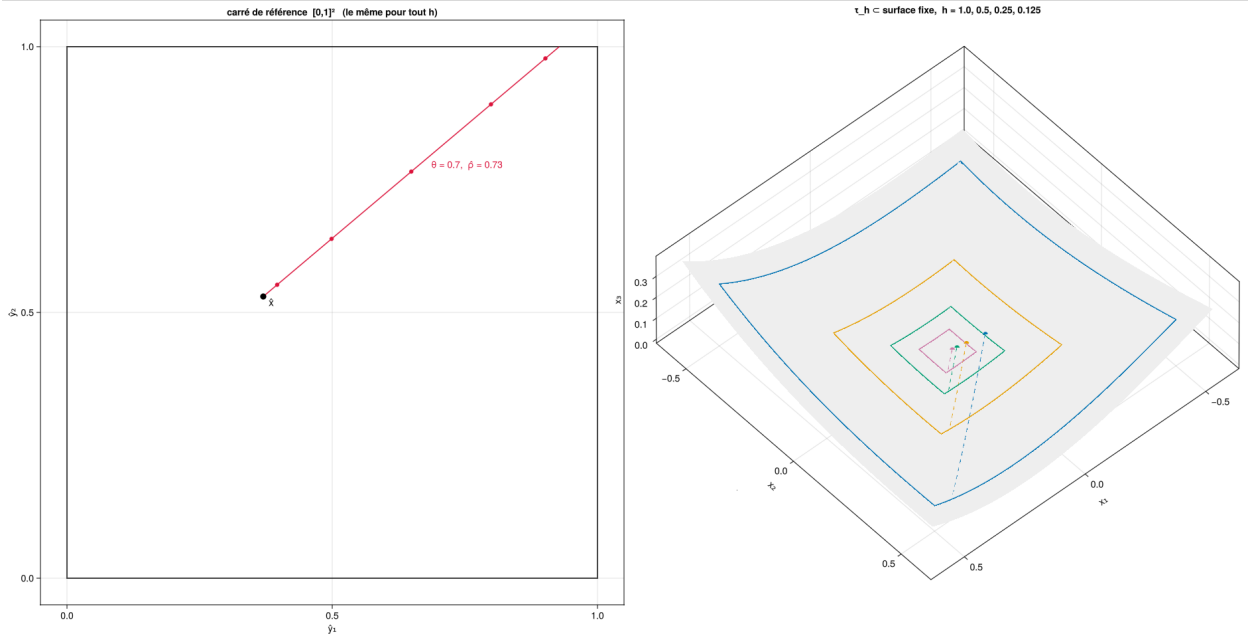


Figure 6: Visualization of the element and its size h .